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The Making of a Grand Canyon

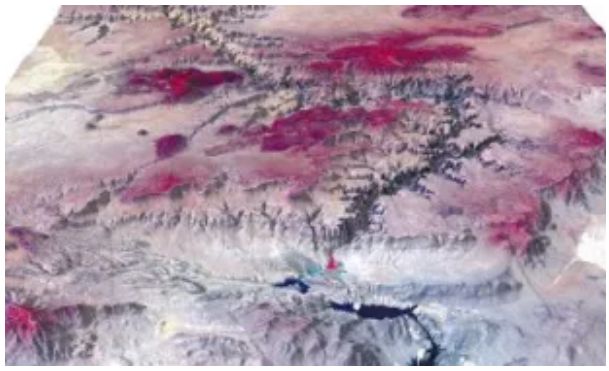
Carving this beloved hole in the ground may not have been such a long-term project

By Sid Perkins

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It would be difficult to find anywhere else in the world a spot yielding so much subject matter for the contemplation of the geologist; certainly there is none situated in the midst of such dramatic and inspiring surroundings.

—Clarence E. Dutton, 1882



A simulated view of the southwestern edge of the Colorado Plateau from space. In the foreground, viewed from the west, the canyon-carving Colorado River flows into the upper reaches of Lake Mead.

STEPHEN J. REYNOLDS/ARIZONA STATE UNIVERSITY

As many as 5 million people each year visit Grand Canyon National Park. Most go for the majestic view, but geologists go to the canyon—a gaping chasm more than 275 miles long, up to 18 miles wide, and in places more than a mile deep—to unravel its untold tale of erosion writ large.

What they're finding is that this epic tale could be, in fact, a short story, geologically speaking. Painted in a thousand earth tones, the canyon's strata eloquently proclaim a history billions of years in the making. Thick bands of limestone speak of lengthy eras spent beneath broad, shallow seas or still, clear lakes. Beds of lava and ash recount episodes of widespread volcanic

activity. Other layers of rock show that this grand swath of real estate has also been home to coastal mudflats and windblown sands.

Although the ancient layers of granite in the deepest part of the canyon are 1.7 billion years old, scientists agree that the gorge itself is nowhere near that age. About 75 years ago, geologists proposed that the Grand Canyon could be as little as 40 million years old. Now, however, evidence is mounting that the canyon is much younger still.

In fact, research presented last June at a conference devoted to the origin of the gorge—the first such meeting in more than 35 years—suggests that substantial portions of the eastern Grand Canyon are geological youngsters, having been eroded only within the past million years.

Grand gash

The Grand Canyon is a gash that's been carved into the southwestern edge of the Colorado Plateau, a 150,000-square-mile, kidney-shaped region that covers large portions of Arizona, New Mexico, Utah, and Colorado. From the rim of the gorge, tourists can often see and sometimes even hear the Colorado River as it makes its way to the Gulf of California.

Until the early 1900s, geologists held that the river had maintained its present southwesterly course across northern Arizona throughout the life of the canyon. According to this view, tectonic forces lifted the Colorado Plateau while sediments carried by the river and its tributaries gradually chewed their way downward to form the gorge.

In the 1930s, however, geologists began to accumulate clues indicating the relative youth of the canyon. Scientists later used radioactive dating to determine the ages of layers of the lava and ash that have repeatedly blanketed the area. In the early 1960s, gravel beds atop the plateau and sediments downstream at the mouth of the Colorado

River supplied new data about the age and configuration of the area's ancient river systems.

In 1964, geologists gathered in Flagstaff, Ariz., to try to assemble these disparate findings into a comprehensive theory of how the canyon came to be. Richard A. Young, now a geologist at the State University of New York in Geneseo, attended the meeting as a graduate student.

Young says most of the ideas that came out of that meeting have survived, but new research continues to fill in the blanks and pose additional tantalizing questions about the early days of the Grand Canyon.

Youthful end

The lower end of the Grand Canyon is one place that tells researchers just how youthful the mile-deep gorge might be. At that part of the gorge, and about halfway between the surface of the Colorado River and the canyon's rim, lies a band of limestone within sediments that geologists call the Muddy Creek Formation. This layer of white rock, the bottom portions of which gradually blend into the silts and sands of the layer below, is several hundred feet thick and appears on both sides of the canyon.

Like other sedimentary rocks, the Muddy Creek Formation is a testament to the environmental conditions in which it formed. Young says the formation's limestone was deposited by types of algae that thrive only in clear, freshwater lakes, suggesting that such a lake once covered the area.

The eventual disappearance of the silt and sand from the limestone indicates that the streams that fed the lake gradually began to dry up and carry less sediment. Perhaps, Young notes, this ancient lake in its later years was fed only by springs.

Radioactive dating of the many layers of ash in the limestone shows that the lake straddled the area between 6 million and 11 million years ago. This is a critical piece of information, Young says, because it places an age limit on the Grand Canyon. If there was a clear lake with no river delta here 6 million years ago, the muddy Colorado River couldn't have been carving the gorge before that time.

Young says this scenario also suggests that the portions of the present-day Colorado River above and below the canyon may not always have been connected. Support for that idea comes from the river delta at the Gulf of California. About 5 million years ago, the sediments dumped by the river suddenly began to include microscopic fossils eroded from the Colorado Plateau that are found nowhere else.

The most likely explanation for the abrupt appearance of these fossils, Young says, is that a west-flowing tributary of the ancestral lower Colorado River began to carve a small valley eastward into the edge of the Colorado Plateau. The upper portion of the tributary eventually merged with the ancestral upper Colorado River and its tributaries to form a single river system.

The result would have been a strengthened torrent of water that could carve its way through rock at a faster clip than ever before. Talk of this scenario began in earnest at the 1964 meeting.

"Fifty years ago, geologists didn't realize how fast erosion could occur," Young says. "When there's a depression in the rock and the river flows through, it can erode incredibly rapidly."

At this summer's meeting, Young and Francis M. Gough presented data that suggest just how efficiently flowing water can cut into the terrain. About 50 miles south of the Grand Canyon, a small river called West Clear Creek is chewing its way eastward through rocks

similar to those that the Colorado began to erode millions of years ago.

Even though the river is only 30 miles long and drains an area of just 350 square miles, Young and Gough report that it has carved a canyon 20 miles long in just a few million years. At its mouth, the gorge is the same width and about two-thirds the depth of the Grand Canyon.

New focus

New data from yet another canyon also show just how fast a river can slash through rock. Robert F. Biek and Grant C. Willis of the Utah Geological Survey in Salt Lake City studied erosion rates along the Virgin River in southwestern Utah, just north of the Grand Canyon. They focused on points both upstream and downstream of the Hurricane Fault, which runs north to south and cuts directly across the river. Using radioactive-dating methods on samples from lava flows and ash beds, the researchers determined the amount of fault movement as well as the rate at which the river chewed downward.

Biek and Willis found that over the past million years, the west side of the fault has dropped about 1,300 feet relative to the east side—an average of about 16 inches every 1,000 years. On the east side of the fault, the river has eaten away 16 inches of rock every millennium. In other words, the river has chewed downward on the higher side of the fault to match the river level on the lower side.

In a smaller canyon near the Utah-Colorado border, measurements by Biek and Willis showed that the Colorado River has cut through about 360 feet of rock in the past 620,000 years. Downstream in the Grand Canyon, where the Colorado carries much more water and sediment, rates of erosion are likely much higher. It's possible, Young says, that the river carved as much as 1,000 feet of the canyon's depth in the past million years.

Circumstantial evidence

While many presentations at this year's meeting filled in gaps in the 1964 hypothesis of rapid formation of the Grand Canyon, some posed alternative scenarios that could revamp views about the early history of the gorge.

For example, circumstantial evidence is mounting that erosion of the gorge could have been started by the floodwaters of a small lake that stood near where the eastern Grand Canyon sits today, says George H. Billingsley, a geologist with the U.S. Geological Survey in Flagstaff.

Among other evidence presented at the meeting, the concentration of strontium in a band of sediments found along the lower Colorado River—rocks known as the Bouse Formation—suggests that a lake fed by the ancestral upper Colorado River began to overflow about 5.5 million years ago. After the water broke through the edge of the basin, it spilled across the Colorado Plateau and began to gouge the Grand Canyon, says Norman Meek, a geographer at California State University in San Bernardino. Over the next 1.5 million years, Meek suggests, the resulting flow carved much of the Grand Canyon and other gorges in eastern Arizona.

Large amounts of roundstone gravel located in the upper portions of the Muddy Creek Formation sediments near the western end of the Grand Canyon provide more support for the overflowing lake scenario. These sediments are now 600 feet thick in places and could be thicker in unexposed areas, says Scott Lundstrom, a Denver-based geologist with the U.S. Geological Survey.

The gravel probably was deposited between 4.75 million and 6.9 million years ago, according to radioactive dating of volcanic deposits that sandwich similar layers of gravel just a few miles away. That period corresponds to the era when Meek proposes the lake spilled

over, releasing what Lundstrom estimates could have been around 1,000 cubic miles of water.

Despite the accumulating data, Billingsley notes that geologists may never be able to verify the early history of their favorite hole in the ground.

“Most of the evidence is gone, because the canyon swallowed the clues to its early history as it grew wider and deeper,” Billingsley says. “It’s a puzzle with too many pieces missing.”

Discovering new clues to the life and times of the Grand Canyon will take geological detective work of the highest order from the few researchers now investigating the subject. The hints are likely scattered to the far corners of the Colorado Plateau and beyond.

By calling attention to new data about the canyon at a forum such as this summer’s meeting, which was held on the southern rim of the canyon, Billingsley and his colleagues are trying to get “a whole new generation of students excited,” he says.

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